Chapter 15, Problem 46P

Problem

Given the following functions

$x(t) = 2\delta(t)$, $y(t) = 4u(t)$, $z(t) = e{-}2tu(t)$,

evaluate the following convolution operations.

(a) $x(t) * y(t)$

(b) $x(t) * z(t)$

(c) $y(t) * z(t)$

(d) $y(t) * [y(t) + z(t)]$

Step-by-step solution

Step 1 of 4

(a)

Evalute $x(t)*y(t)$.

Substitute $2\delta(t)$ for $x(t)$ and $4u(t)$ for $y(t)$.

$x(t)*y(t)=2\delta(t)*[4u(t)]$ …… (1)

Apply Convolution property to this expression.

Refer to equation 15.78 in the textbook.

$$F_1(s)F_2(s)=\mathcal{L}[f_1(t)*f_2(t)]$$

Refer to Table 15.2 for Laplace transform pairs.

$$\mathcal{L}[\delta(t)]=1$$

$$\mathcal{L}[u(t)]=\frac{1}{s}$$

$$\mathcal{L}[e^{-at}]=\frac{1}{s+a}$$

$$\mathcal{L}[tu(t)]=\frac{1}{s^2}$$

Therefore,

$$\begin{aligned}2\delta(t)*[4u(t)]&=\mathcal{L}^{-1}\{(\mathcal{L}[2\delta(t)])(\mathcal{L}[4u(t)])\}\\&=\mathcal{L}^{-1}\left\{2(1)\left(\frac{4}{s}\right)\right\}\\&=8\mathcal{L}^{-1}\left\{\frac{1}{s}\right\}\\&=8u(t)\end{aligned}$$

Therefore, the required expression for $x(t)*y(t)$ is $\boxed{8u(t)}$.

Step 2 of 4

(b)

Evalute $x(t)*z(t)$.

Substitute $2\delta(t)$ for $x(t)$ and $e^{-2t}u(t)$ for $z(t)$.

$x(t)*z(t)=2\delta(t)*\left[e^{-2t}u(t)\right]$ ...... (2)

Apply Convolution property to this expression.

$$\begin{aligned}2\delta(t)*\left[e^{-2t}u(t)\right]&=L^{-1}\left\{\left(L\left[2\delta(t)\right]\right)\left(L\left[e^{-2t}u(t)\right]\right)\right\}\\&=L^{-1}\left\{(2)\left(\frac{1}{s+2}\right)\right\}\\&=2L^{-1}\left\{\frac{1}{s+2}\right\}\\&=2e^{-2t}u(t)\end{aligned}$$

Therefore, the required expression for $x(t)*z(t)$ is $\boxed{2e^{-2t}u(t)}$.

Step 3 of 4

(c)

Evalute $y(t)*z(t)$.

Substitute $4u(t)$ for $y(t)$ and $e^{-2t}u(t)$ for $z(t)$.

$y(t)*z(t)=4u(t)*\left[e^{-2t}u(t)\right]$ ...... (3)

Apply Convolution property to this expression.

$$\begin{aligned}4u(t)*\left[e^{-2t}u(t)\right]&=L^{-1}\left\{\left(L\left[4u(t)\right]\right)\left(L\left[e^{-2t}u(t)\right]\right)\right\}\\&=L^{-1}\left\{\frac{4}{s}\left(\frac{1}{s+2}\right)\right\}\\&=L^{-1}\left\{2\left(\frac{1}{s}-\frac{1}{s+2}\right)\right\}\end{aligned}$$

$$\begin{aligned}4u(t)*\left[e^{-2t}u(t)\right]&=2L^{-1}\left\{\frac{1}{s}-\frac{1}{s+2}\right\}\\&=2L^{-1}\left\{\frac{1}{s}\right\}-2L^{-1}\left\{\frac{1}{s+2}\right\}\\&=2u(t)-2e^{-2t}u(t)\\&=2\left(1-e^{-2t}\right)u(t)\end{aligned}$$

Therefore, the required expression for $y(t)*z(t)$ is $\boxed{2\left(1-e^{-2t}\right)u(t)}$.

Step 4 of 4

(d)

Evalute $y(t)*[y(t)+z(t)]$.

Substitute $4u(t)$ for $y(t)$ and $e^{-2t}u(t)$ for $z(t)$.

$$y(t)*[y(t)+z(t)]=4u(t)*[4u(t)+e^{-2t}u(t)] \quad \ldots\ldots (4)$$

Apply Convolution property to the expression (4).

$$\begin{aligned}4u(t)*[4u(t)+e^{-2t}u(t)] &= L^{-1}\{(L[4u(t)])(L[4u(t)+e^{-2t}u(t)])\} \\ &= L^{-1}\left\{\frac{4}{s}\left(\frac{4}{s}+\frac{1}{s+2}\right)\right\} \\ &= L^{-1}\left\{\frac{16}{s^2}+\frac{4}{s(s+2)}\right\} \\ &= L^{-1}\left\{16\frac{1}{s^2}+\frac{2}{s}-\frac{2}{s+2}\right\}\end{aligned}$$

Simplify further.

$$\begin{aligned}4u(t)*[4u(t)+e^{-2t}u(t)] &= 16L^{-1}\left\{\frac{1}{s^2}\right\}+2L^{-1}\left\{\frac{1}{s}\right\}-2L^{-1}\left\{\frac{1}{s+2}\right\} \\ &= 16tu(t)+2u(t)-2e^{-2t}u(t) \\ &= 2(8t+1-e^{-2t})u(t)\end{aligned}$$

Therefore, the required expression for $y(t)*[y(t)+z(t)]$ is $\boxed{2(8t+1-e^{-2t})u(t)}$.